QP CODE: 24900137

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024

First Semester

Multi-Disciplinary Course - MG1MDCHIS101 - SCIENCE AND SOCIETY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Part A Short Answer Questions Answer any 5 out of 8 questions. Each question carries 2 marks

1) Epistemology- branch of philosophy that studies the nature, scope, and limits of knowledge- examines the sources, methods, and validity of knowledge, focusing on how we know what we know.
   (Aligns with CO1: Understand the epistemological developments in modern science.)
2) "Cogito, ergo sum" - Latin phrase- "I think, therefore I am”- René Descartes- thinking is proof of one's existence and consciousness.
   (Aligns with CO1: Understand the epistemological developments in modern science.)
3) "Survival of the fittest" -evolutionary theory- organisms best adapted to their environment are more likely to survive and reproduce- natural selection.
   (Aligns with CO2: Analyse the non-European roots of scientific traditions.)
4) Isaac Newton- Philosophia Naturalis Principia Mathematica- the laws of motion and universal gravitation- foundation for classical mechanics- role in revolutionising scientific understanding.
   (Aligns with CO1: Understand the epistemological developments in modern science.)
5) A priori knowledge- knowledge independent of experience- known through reason alone- considered universally valid without empirical evidence.
   (Aligns with CO1: Understand the epistemological developments in modern science.)
6) Radiocarbon dating- method for determining the age of an object containing organic material-measuring the amount of carbon-14- based on the principle that carbon-14 decays at a known rate over time.

(Aligns with CO2: Analyse the non-European roots of scientific traditions.)

7) 'Paradigm Shift'- fundamental change in the underlying assumptions or approach within a particular field of science or discipline- occurs when discoveries or ideas challenge the established framework, leading to a significant transformation in thought and practice.
   (Aligns with CO1: Understand the epistemological developments in modern science.)
8) 'Indigenous Knowledge'- traditional, local knowledge systems- developed through their long-standing interaction with the environment- encompasses practices, beliefs, and skills passed down through generations, often rooted in cultural and ecological contexts.
   (Aligns with CO2: Analyse the non-European roots of scientific traditions.)

Part B Short Essay Questions Answer any 4 out of 7 questions. Each question carries 5 marks.

9) Descartes' methodical doubt- I think, therefore I am- promoting scepticism as a tool for uncovering the truth- doubting everything that could be doubted- establishes a foundation of specific knowledge based solely on reason- shifted the focus of inquiry from reliance on tradition or authority to individual rational investigation- encouraged future scientists to question assumptions, emphasise empirical evidence, and adopt a systematic approach to discovering knowledge- Descartes helped shape the scientific method, prioritising doubt, experimentation, and evidence-based conclusions.
   (Aligns with CO1: Understand the epistemological developments in modern science.)
10) Charles Darwin- theory of natural selection-organisms better suited to their environment are more likely to survive and reproduce- advantageous traits to future generations- tiny changes accumulate leading to the evolution of new species- natural selection operates on variation within populations, with environmental pressures determining favourable characteristics- forms the cornerstone of evolutionary biology and provides a scientific explanation for the diversity of life on Earth.
   (Aligns with CO2: Analyse the non-European roots of scientific traditions.)
11) The institutionalisation of scientific practice in Europe- Renaissance and Enlightenment- scientific inquiry shifted from primarily pursuing individuals or small groups to becoming a more formalised and institutionalised activity- establishment of scientific societies like the Royal Society of London in 1660- platforms for collaboration, knowledge sharing, and the standardisation of scientific methods- Universities - incorporating science into their curricula- institutionalisation helped legitimise science as a professional discipline and facilitated its rapid advancement.

(Aligns with CO1: Understand the epistemological, methodological, and institutional developments in modern science.)

12) The material milieu of science in the Arab world during its Golden Age (8th to 14th centuries)- shaped by a combination of intellectual, cultural, and material conditions that fostered scientific development- rise of the Islamic caliphates-establishment of institutions like the House of Wisdom in Baghdad- translation of Greek, Persian, and Indian texts into Arabic-significant advancements in fields like mathematics, astronomy, medicine, and chemistry- role of trade, wealth from the empire, and a network of scholars in the growth of scientific inquiry.
(Aligns with CO2: Analyse the non-European roots of scientific traditions.)

13) Europe owes a significant debt to Chinese science-printing, gunpowder, and the compass- introduction of these technologies and the shaping of Renaissance and subsequent European advancements- *relevant examples should be cited*
(Aligns with CO2: Analyse the non-European roots of scientific traditions.)

14) Postmodernism- challenging the idea of objective, universal truths in science-scientific knowledge is socially constructed and shaped by cultural, political, and historical contexts - critiques the traditional view of science as an impartial pursuit of truth, emphasising instead that power dynamics, ideologies, and the interests of dominant groups influence scientific practices- focus on the subjectivity involved in scientific inquiry, calling for more inclusive and diverse perspectives in scientific research- rethinking of scientific authority and a more critical view of the assumptions underlying scientific methodologies.
(Aligns with CO3: Assess how science viewed social structures and institutions in the Western world and colonies.)

15) KSSP and TLC- aimed to eradicate illiteracy- KSSP- focus on integrating scientific temper with education- grassroots approach to promote literacy-community-based initiatives, including workshops, awareness campaigns, and disseminating educational materials-emphasis on the importance of science and rational thinking- questioning of myths and superstitions- role in achieving complete literacy
(Aligns with CO4: Generate scientific temper among students to counter unscientific and mythicised interpretations of the past.)

Part C Essay Questions Answer any 2 out of 4 questions. Each question carries 10 marks

16) Darwin- origin and dynamics of species - foundational to modern biology- theory of evolution by natural selection- On the Origin of Species (1859)-natural selection- beneficial traits in a population- voyage on the HMS Beagle-Galapagos Islands-insights into how species adapt to their environments- idea of common descent- challenged traditional views of creation- biodiversity and the interconnectedness of life forms- modern genetics- study of life into an evidence-

based science-evolutionary biology, ecology, and genetics, which continue to explore the complex dynamics of life on Earth.

(Aligns with CO2: Analyse the non-European roots of scientific traditions and CO1: Understand the epistemological developments in modern science.)

17) Astronomical and mathematical knowledge in ancient India-advancements in understanding the cosmos- Rigveda and Yajurveda- references to celestial bodies and their movements- Aryabhata- concept of a heliocentric universe-value of pi - Earth rotates on its axis-Indian number system- concept of zero-decimal place value and algebra concepts-Surya Siddhanta- calculations of planetary movements and eclipses- Nalanda University

(Aligns with CO2: Analyse the non-European roots of scientific traditions.)

18) KSSP- focus on mass science education- broader form of political activism- new democratic initiatives in Kerala-promoting scientific temper and rational thinking- addressing superstition and unscientific beliefs-TLC- scientific education- empowering the masses and reduce illiteracy-transformative power of knowledge and scientific reasoning to improve social conditions- expanded its activism to address broader issues such as social justice, environmental protection, and political empowerment- involvement in political campaigns that aimed to improve people's lives through education and activism- advocacy of progressive policies-using science to challenge social and political structures-comprehensive approach, blending scientific education with grassroots political engagement

(Aligns with CO4: Generate scientific temper among students to counter unscientific and mythicised interpretations of the past.)

19) Science-tool of both industrial capitalism and imperialism- development of modern economies and global power dynamics- drive technological innovations, improve production efficiency, and enhance profit margins- engineering, chemistry, and agriculture-mass production of goods- rise of industrial capitalism- scientific advancements in machinery, transportation, and communication-expansion of capitalist economies- growth of industrial giants and global trade networks- role played in the expansion of imperialism-justification and sustenance of colonial domination by providing "rational" explanations- scientific exploration to map territories, exploit resources, and develop new control over colonised nations-science of race- imperialism fuelled scientific exploration and discovery- knowledge about local resources, biodiversity, and geography to benefit their economies.

(Aligns with CO3: Assess how science looked at social structures and institutions in the Western world and colonies.)